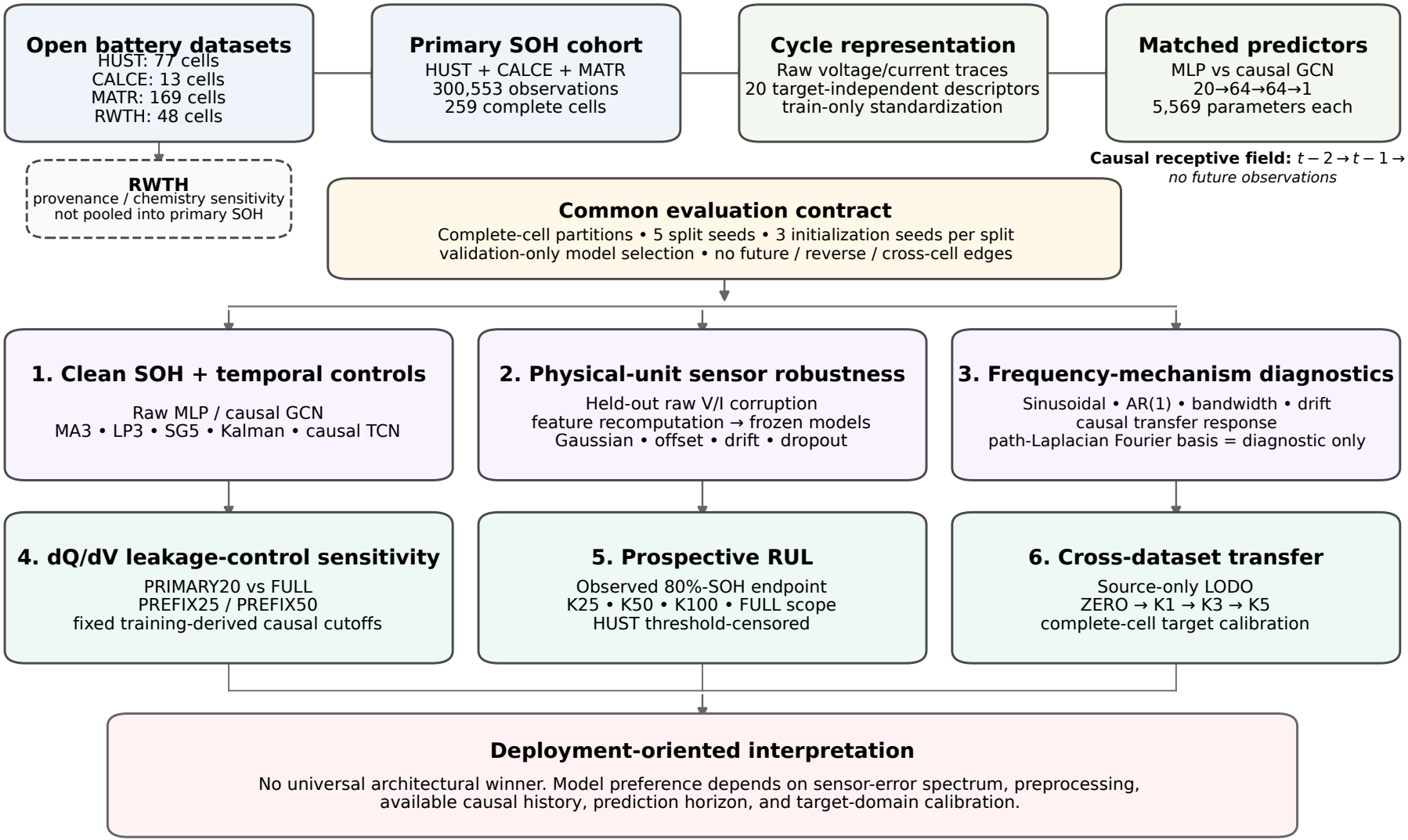
Open battery datasets
HUST: 77 cells
CALCE: 13 cells
MATR: 169 cells
RWTH: 48 cells
Primary SOH cohort
HUST + CALCE + MATR
300,553 observations
259 complete cells
Cycle representation
Raw voltage/current traces
20 target-independent descriptors
train-only standardization
Matched predictors
MLP vs causal GCN
20→64→64→1
5,569 parameters each
Causal receptive field: $t-2 \rightarrow t-1 \rightarrow t$
no future observations
RWTH
provenance / chemistry sensitivity
not pooled into primary SOH
Common evaluation contract
Complete-cell partitions • 5 split seeds • 3 initialization seeds per split
validation-only model selection • no future / reverse / cross-cell edges
1. Clean SOH + temporal controls
Raw MLP / causal GCN
MA3 • LP3 • SG5 • Kalman • causal TCN
2. Physical-unit sensor robustness
Held-out raw V/I corruption
feature recomputation → frozen models
Gaussian • offset • drift • dropout
3. Frequency-mechanism diagnostics
Sinusoidal • AR(1) • bandwidth • drift
causal transfer response
path-Laplacian Fourier basis = diagnostic only
4. dQ/dV leakage-control sensitivity
PRIMARY20 vs FULL
PREFIX25 / PREFIX50
fixed training-derived causal cutoffs
5. Prospective RUL
Observed 80%-SOH endpoint
K25 • K50 • K100 • FULL scope
HUST threshold-censored
6. Cross-dataset transfer
Source-only LODO
ZERO → K1 → K3 → K5
complete-cell target calibration
Deployment-oriented interpretation
No universal architectural winner. Model preference depends on sensor-error spectrum, preprocessing,
available causal history, prediction horizon, and target-domain calibration.